

Experiment 'p_6_2_j_has_duplicates_seq' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_j_has_duplicates_seq

Natural language statement: null

Method signature: p_6_2_j_has_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)

Ensures

- `has_duplicates == (exists i, j :: 0 <= i < j < |inputs| && inputs[i] == inputs[j])`

Requires

- `|inputs| >= 2`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_j_has_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)`

The method should respect the following contract:

`requires |inputs| >= 2, ensures has_duplicates == (exists i, j :: 0 <= i < j < |inputs|
↪ && inputs[i] == inputs[j])`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnOAPrdhF5M6fhDjlx0tdfFcGdO4l

Model: gpt-5.5-2026-04-23

Created at: 1780663341

Finish reason: stop

Usage: 308 tokens in, and 923 tokens out

```
{  
  if exists i, j :: 0 <= i < j < |inputs| && inputs[i] == inputs[j] {  
    has_duplicates := true;  
  } else {  
    has_duplicates := false;  
  }  
}
```

Final Program

```
method p_6_2_j_has_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)  
  requires |inputs| >= 2  
  ensures has_duplicates == (exists i, j :: 0 <= i < j < |inputs| && inputs[i] ==  
    ↪ inputs[j])  
{  
  if exists i, j :: 0 <= i < j < |inputs| && inputs[i] == inputs[j] {  
    has_duplicates := true;  
  } else {
```

```
    has_duplicates := false;  
  }  
}
```

Total Token Usage

Input tokens: 308

Output tokens: 923

Reasoning tokens: 847

Sum of ‘total tokens’: 1231

Experiment Timings

Overall Experiment started at 1780663340844, ended at 1780663383996, lasting 43152ms (43.15 seconds)

Iteration #1 started at 1780663340844, ended at 1780663383996, lasting 43152ms (43.15 seconds)